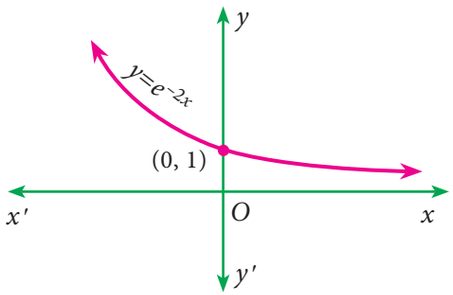
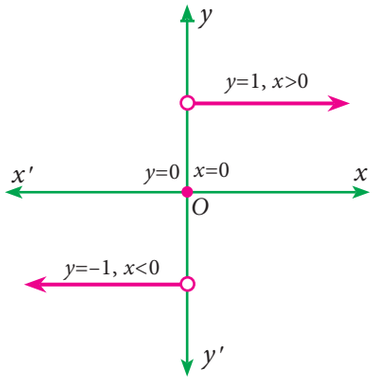
(v)



(vi)



Exercise 5.2

1.(i) $\frac{10}{3}$ (ii) 0 (iii) $\frac{1}{2}$ (iv) 1

(v) $\frac{15}{16}a^{-\frac{1}{24}}$ (vi) 9

2. $a = \pm 1$ 3. $n = 7$ 4. $\frac{28}{5}$ 5. $f(-2) = 0$

Exercise 5.3

1.(a) Not a continuous function at $x = 2$ (b) continuous function at $x = 3$

Exercise 5.4

(i) $2x$ (ii) $-e^{-x}$ (iii) $\frac{1}{x+1}$

Exercise 5.5

1.(i) $12x^3 - 6x^2 + 1$ (ii) $\frac{-20}{x^5} + \frac{6}{x^4} - \frac{5}{x^2}$ (iii) $\frac{1}{2\sqrt{x}} - \frac{1}{3\sqrt[3]{x^4}} + e^x$

(iv) $-\frac{3+x^2}{x^2}$ (v) $x^2e^x(x+3)$ (vi) $3x^2 - 4x - 1$

(vii) $4x^3 - 3\cos x - \sin x$ (viii) $1 - \frac{1}{x^2}$

2.(i) $\frac{xe^x}{(1+x)^2}$ (ii) $\frac{2(1-x^2)}{(x^2-x+1)^2}$ (iii) $\frac{e^x}{(1+e^x)^2}$

3.(i) $x\cos x + \sin x$ (ii) $e^x(\sin x + \cos x)$

(iii) $e^x\left(1 + \frac{1}{x} + x + \log x\right)$ (iv) $\cos 2x$ (v) $e^x x^2(x+3)$

4.(i) $\sin 2x$ (ii) $-\sin 2x$ (iii) $\frac{-3}{2}\cos x \sin 2x$ (iv) $\frac{x}{\sqrt{1+x^2}}$

(v) $n(ax^2+bx+c)^{n-1}(2ax+b)$ (vi) $2x(\cos(x^2))$ (vii) $\frac{-x}{(1+x^2)\sqrt{1+x^2}}$

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